

How they get their numbers: Both programs use plain forward Euler integration — step S/I/R (and E for SEIR) forward in fixed $dt=0.001$ -day increments, recomputing each derivative from the current state each time, ~60,000 steps for SIR ($t_{\text{Max}}=60$) and ~100,000 for SEIR ($t_{\text{Max}}=100$). SIR uses $\beta=0.5$, $\gamma=0.2$, $N=10000$, $S_0=9900$, $I_0=10$, $R_0=0$. SEIR uses the same β/γ plus $\sigma=0.2$ (5-day incubation), $\mu=0$.

How it compares to the paper: For $\beta=0.5$, $\gamma=0.2$ ($R_0=2.5$), the paper's Table 1 reports peak $I=2339$ on day 24.4, final outbreak size 8916. The code's actual output (confirmed from the spreadsheet) is peak $I=2285.14$ on day 24.6, final size ≈ 8895.8 — off by about 2%. The cause: the paper's own text says $N_1=9900$, but its own Appendix code actually uses $S_0=N-10=9990$ (the paper contradicts itself). Your code followed the text (9900) while still dividing by $N=10000$ in the transmission term, so $S_0+I_0+R_0=9910$ instead of 10000 — a ~90-person population mismatch that slightly dilutes transmission. I tested this directly: rerunning the identical Euler code with $S_0=9990$ instead of 9900 gives peak $I=2338.99$ on day 24.36, final size 8916.76 — matching the paper almost exactly. That confirms the Euler method itself (at $dt=0.001$) isn't the issue; it reproduces the paper's Runge-Kutta results essentially perfectly once the population is consistent.

Separately, the SEIR file doesn't have a paper value to compare against at all — this PDF only covers SIR (plus an SIR+vital-dynamics extension). The SEIR structure and R_0 formula the code uses are from a different paper you read earlier (Blackwood & Childs), so there's no Table 1-style benchmark here to check it against.